In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

sns.set_style("darkgrid")
print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

In [11]:
```
# loading the CSV file into a dataframe
df = pd.read_csv('race_data.csv')
# defining horses, cuz we gonna use this a lot
horses = ['Shadowfax', 'Iron Duke', 'Morningstar', 'Red Tide', 'Gallant F
print("Shape of data (rows, columns):", df.shape)
print("\nFirst 5 rows:")
df.head()
```

```
Shape of data (rows, columns): (500, 10)

First 5 rows:
```

Out[11]:

| | race_id | Shadowfax | Iron Duke | Morningstar | Red Tide | Gallant Fox | Blue Streak | Copper Prince |
|---|---|---|---|---|---|---|---|---|
| **0** | 0 | 67.5041 | 65.6016 | 70.4587 | 73.1363 | 64.2286 | 67.2475 | 75.0616 |
| **1** | 1 | 66.6415 | 66.0323 | 71.0060 | 72.6026 | 71.0442 | 76.8610 | 76.7373 |
| **2** | 2 | 67.6772 | 65.7881 | 71.0019 | 69.9481 | 70.1585 | 69.5850 | 80.5973 |
| **3** | 3 | 65.5535 | 67.2001 | 69.5425 | 71.2678 | 72.2864 | 73.9726 | 85.5591 |
| **4** | 4 | 65.3339 | 66.1231 | 69.8925 | 73.7591 | 70.4077 | 68.9783 | 70.5561 |

In [12]:
```
# --- Win Counts ---
print("How many times each horse won out of 500 races:\n")
print(df['winner'].value_counts())

print("\nAs percentages(empirical win probability):\n")
print((df['winner'].value_counts(normalize=True) * 100).round(2).astype(s
```

```
How many times each horse won out of 500 races:

winner
Shadowfax        178
Morningstar      164
Iron Duke         74
Gallant Fox       28
Red Tide          27
Blue Streak       14
Copper Prince     10
Last Chance        5
Name: count, dtype: int64

As percentages(empirical win probability):

winner
Shadowfax        35.6%
Morningstar      32.8%
Iron Duke        14.8%
Gallant Fox       5.6%
Red Tide          5.4%
Blue Streak       2.8%
Copper Prince     2.0%
Last Chance       1.0%
Name: proportion, dtype: object
```

In [13]:
```
# --- Finishing Times Summary ---
print("Average finishing time per horse (lower = faster):\n")
print(df[horses].mean().round(2).sort_values())

print("\nStandard deviation (how consistent each horse is):\n")
print(df[horses].std().round(2).sort_values())
```

```
Average finishing time per horse (lower = faster):

Shadowfax        66.66
Morningstar      67.32
Iron Duke        68.06
Red Tide         70.15
Gallant Fox      70.88
Blue Streak      72.33
Copper Prince    74.39
Last Chance      77.59
dtype: float64

Standard deviation (how consistent each horse is):

Iron Duke        2.33
Shadowfax        2.59
Red Tide         3.17
Gallant Fox      3.50
Blue Streak      4.05
Morningstar      4.13
Copper Prince    4.84
Last Chance      6.38
dtype: float64
```

In [15]:
```
# Visualise the Distributions

fig, axes = plt.subplots(2, 4, figsize=(18, 8))  # 2 rows, 4 columns of c
```

```
axes = axes.flatten()  # makes it easier to loop through

for i, horse in enumerate(horses):
    ax = axes[i]

    # plot a histogram of the horse's finishing times
    ax.hist(df[horse], bins=30, color='steelblue', edgecolor='white',
            alpha=0.7, density=True, label='Actual data')

    # fit a normal distribution to the data
    mu, std = stats.norm.fit(df[horse])  # mu = mean, std = standard devi

    # plot the fitted normal curve on top
    x = np.linspace(df[horse].min(), df[horse].max(), 100)
    ax.plot(x, stats.norm.pdf(x, mu, std), 'red', linewidth=2, label='Nor

    # labels
    ax.set_title(f'{horse}\nμ={mu:.2f}s, σ={std:.2f}s', fontsize=11)
    ax.set_xlabel('Finish Time (seconds)')
    ax.set_ylabel('Density')
    ax.legend(fontsize=8)

plt.suptitle('Finishing Time Distributions — All 8 Horses',
             fontsize=16, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```


In [17]:
```
# Monte Carlo Simulation

# first, store each horse's mean and std from the real data
means = df[horses].mean()   # average finishing time per horse
stds  = df[horses].std()    # standard deviation per horse

# number of races to simulate — more = more accurate estimates
N = 100_000

# dictionary to count how many times each horse wins
win_counts = {horse: 0 for horse in horses}

# set a random seed so results are reproducible
np.random.seed(42)

print(f"Simulating {N:,} races...")
```

```
for _ in range(N):
    # for each horse, sample a finishing time from its normal distributio
    times = {horse: np.random.normal(means[horse], stds[horse]) for horse

    # winner = horse with the LOWEST finishing time
    winner = min(times, key=times.get)

    # add 1 to that horse's win count
    win_counts[winner] += 1

# convert win counts to probabilities
true_p = {horse: win_counts[horse] / N for horse in horses}

print("Simulation complete!\n")
print("Estimated True Win Probabilities:\n")
for horse, p in sorted(true_p.items(), key=lambda x: -x[1]):
    bar = '' * int(p * 100)
    print(f"{horse:<15} {p*100:5.1f}%  {bar}")
```

```
Simulating 100,000 races...
Simulation complete!

Estimated True Win Probabilities:

Shadowfax        32.4%  ████████████████████████████████
Morningstar      32.1%  ████████████████████████████████
Iron Duke        13.8%  ██████████████
Red Tide          6.8%  ███████
Gallant Fox       5.6%  ██████
Blue Streak       4.1%  ████
Copper Prince     2.8%  ███
Last Chance       2.3%  ███
```

In [20]:

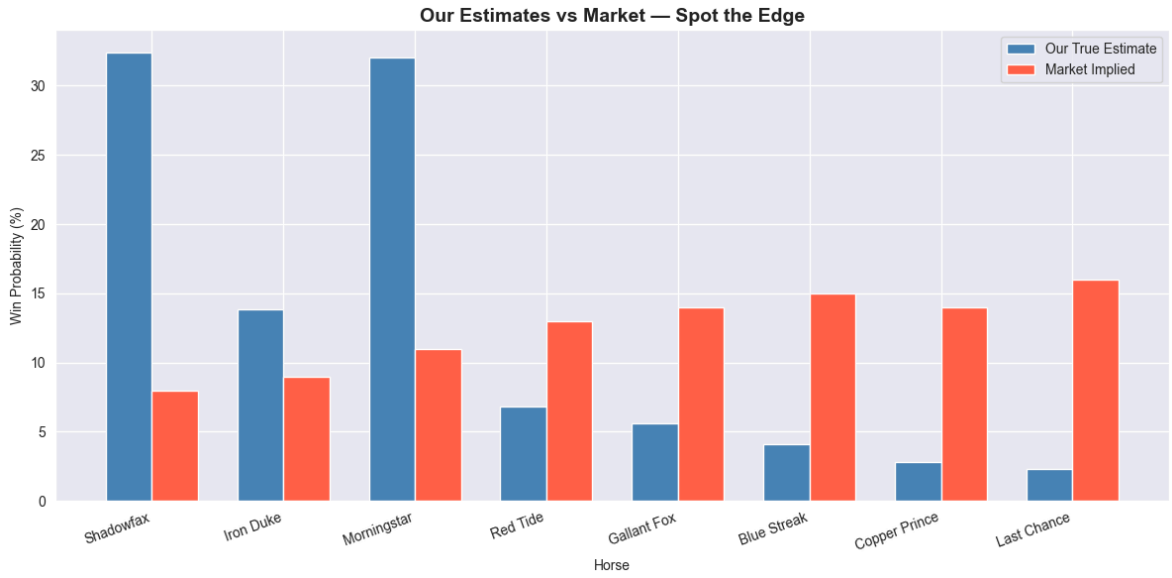
```
# Expected Value Calculation

# market pool fractions (given in the problem)
pool_fractions = {
    'Shadowfax':     0.08,
    'Iron Duke':     0.09,
    'Morningstar':   0.11,
    'Red Tide':      0.13,
    'Gallant Fox':   0.14,
    'Blue Streak':   0.15,
    'Copper Prince': 0.14,
    'Last Chance':   0.16
}

print("=" * 65)
print(f"{'Horse':<15} {'True P':>8} {'Market':>8} {'EV':>8} {'Bet?':>8}")
print("=" * 65)

positive_ev_horses = []  # we'll store the good bets here

for horse in horses:
    p = true_p[horse]              # our estimated true win probability
    f = pool_fractions[horse]      # market's implied probability

    # EV formula from the problem
    # if EV > 0, the bet is profitable in the long run
```

```
    ev = p * (0.85 / f) - 1

    should_bet = "BET" if ev > 0 else "SKIP"

    if ev > 0:
        positive_ev_horses.append(horse)

    print(f"{horse:<15} {p*100:>7.1f}% {f*100:>7.1f}% {ev:>+8.4f}  {shoul

print("=" * 65)
print(f"\nHorses worth betting on: {positive_ev_horses}")
```

```
=================================================================
Horse             True P   Market       EV     Bet?
=================================================================
Shadowfax          32.4%     8.0%  +2.4424  BET
Iron Duke          13.8%     9.0%  +0.3054  BET
Morningstar        32.1%    11.0%  +1.4781  BET
Red Tide            6.8%    13.0%  -0.5523  SKIP
Gallant Fox         5.6%    14.0%  -0.6592  SKIP
Blue Streak         4.1%    15.0%  -0.7682  SKIP
Copper Prince       2.8%    14.0%  -0.8276  SKIP
Last Chance         2.3%    16.0%  -0.8768  SKIP
=================================================================

Horses worth betting on: ['Shadowfax', 'Iron Duke', 'Morningstar']
```

In [21]:

```
# --- Visualise: True P vs Market P ---

true_probs  = [true_p[h] * 100 for h in horses]
market_probs = [pool_fractions[h] * 100 for h in horses]

x = np.arange(len(horses))
width = 0.35

fig, ax = plt.subplots(figsize=(12, 6))
bars1 = ax.bar(x - width/2, true_probs,  width, label='Our True Estimate'
bars2 = ax.bar(x + width/2, market_probs, width, label='Market Implied',

ax.set_xlabel('Horse')
ax.set_ylabel('Win Probability (%)')
ax.set_title('Our Estimates vs Market — Spot the Edge', fontsize=14, font
ax.set_xticks(x)
ax.set_xticklabels(horses, rotation=20, ha='right')
ax.legend()
plt.tight_layout()
plt.show()
```




In [22]:
```
# Kelly Criterion + Final Bet Sizes

BANKROLL = 10_000        # total money we can bet (£10,000)
KELLY_FRACTION = 0.25    # quarter-Kelly — conservative, protects against

print("=" * 60)
print(f"{'Horse':<15} {'Kelly %':>8} {'Stake (£)':>10}")
print("=" * 60)

final_bets = {}
total_staked = 0

for horse in horses:
    p = true_p[horse]
    f = pool_fractions[horse]
    ev = p * (0.85 / f) - 1

    if ev <= 0:
        # negative EV — don't bet
        final_bets[horse] = 0.0
        print(f"{horse:<15} {'SKIP':>8} {'£0.00':>10}")
        continue

    # net odds: if you bet £1 and win, how much profit do you make?
    b = (0.85 / f) - 1

    # full kelly formula: what fraction of bankroll to bet?
    # derived from: maximise expected log(wealth)
    kelly = (p * b - (1 - p)) / b

    # scale down by our fraction (quarter kelly = safer)
    stake = kelly * KELLY_FRACTION * BANKROLL
    stake = round(stake, 2)

    final_bets[horse] = stake
    total_staked += stake

    print(f"{horse:<15} {kelly*100:>7.2f}% {stake:>10.2f}")

print("=" * 60)
```

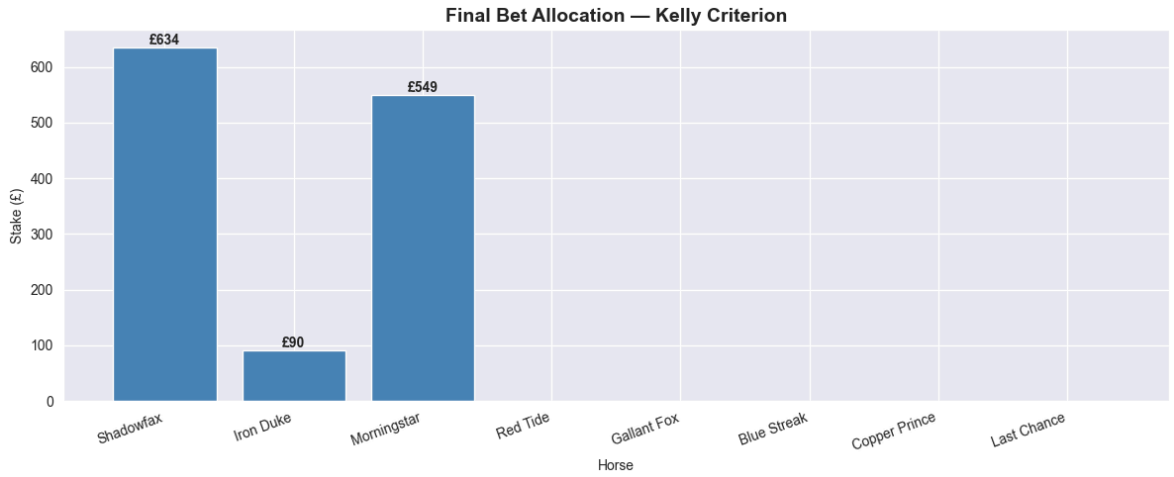
```
print(f"{'TOTAL STAKED':<15} {'':>8} {total_staked:>10.2f}")
print(f"{'KEPT IN HAND':<15} {'':>8} {BANKROLL - total_staked:>10.2f}")
```

```
============================================================
Horse            Kelly %  Stake (£)
============================================================
Shadowfax         25.38%     634.39
Iron Duke          3.62%      90.42
Morningstar       21.97%     549.31
Red Tide            SKIP      £0.00
Gallant Fox         SKIP      £0.00
Blue Streak         SKIP      £0.00
Copper Prince       SKIP      £0.00
Last Chance         SKIP      £0.00
============================================================
TOTAL STAKED                 1274.12
KEPT IN HAND                 8725.88
```

In [26]:

```
# --- Final Answer Summary ---

print("\n" + "=" * 40)
print("         FINAL SUBMISSION")
print("=" * 40)
for horse, stake in final_bets.items():
    status = "✅" if stake > 0 else "  "
    print(f"{status} {horse:<15}  £{stake:.2f}")
print("=" * 40)
print(f"   Total: £{total_staked:.2f} / £{BANKROLL:.2f}")
print("=" * 40)

# --- Bar chart of final bets ---
stakes = [final_bets[h] for h in horses]

plt.figure(figsize=(12, 5))
colors = ['steelblue' if s > 0 else 'lightgrey' for s in stakes]
plt.bar(horses, stakes, color=colors, edgecolor='white')
plt.title('Final Bet Allocation — Kelly Criterion', fontsize=14, fontweig
plt.xlabel('Horse')
plt.ylabel('Stake (£)')
plt.xticks(rotation=20, ha='right')

# add £ labels on top of each bar
for i, (h, s) in enumerate(zip(horses, stakes)):
    if s > 0:
        plt.text(i, s + 5, f'£{s:.0f}', ha='center', fontweight='bold')

plt.tight_layout()
plt.show()
```

```
========================================
        FINAL SUBMISSION
========================================
✅ Shadowfax        £634.39
✅ Iron Duke        £90.42
✅ Morningstar      £549.31
   Red Tide         £0.00
   Gallant Fox      £0.00
   Blue Streak      £0.00
   Copper Prince    £0.00
   Last Chance      £0.00
========================================
   Total: £1274.12 / £10000.00
========================================
```




In [ ]: